



Constrained optimal power flow and optimal TCSC allocation using hybrid cuckoo search and ant lion optimizer

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Abstract This paper works on the solution to Voltage Constrained Optimal Power Flow (VCOPF) problem with the requisite allocation of thyristor-controlled series compensator (TCSC) in IEEE 6-bus and 14-bus transmission network to cut down system power losses and to revamp bus voltage profile. The Newton–Raphson algorithm computes the power flow under normal and overloaded operating conditions. The optimal TCSC location is identified by the Cuckoo Search algorithm (CS) and optimal size is determined using an ant-lion optimizer (ALO). The quadratic fuel cost is chosen as the objective and is subjugated to the equality and inequality constraints. The proposed methodology is validated by contrasting the results to the other hybrid methods such as Fuzzy-Gravitational search algorithm (F-GSA), Improved GSA-Firefly algorithm (IGSA-FA) and Radial basis function neural network-GSA (RBFNN-GSA). The statistical analysis is also carried out for validating the efficacy of the algorithm when compared with other reported methods in the literature. The simulation results obtained on standard test

systems manifest the improved performance of proposed hybrid cuckoo search and ant lion optimizer (CS-ALO) in comparison with the other optimization techniques that have emerged in the recent state-of-the-art-literature.

Keywords TCSC · Optimal allocation · Ant-lion optimizer · Cuckoo search · Optimal power flow

1 Introduction

1.1 Motivation

The optimal power flow (OPF) integrates the power flow with the dispatch to minimize the fuel cost subject to equality and inequality constraints. The non-linearity Newton method appreciably addresses the marginal losses but the linear programming is more useful in determining the boundary limits. While solving the OPF problem, the system safety parameters such as voltage violations and thermal capacity of the lines must be regulated precisely (Charles et al. 2020; Weinhold and Mieth 2020). The OPF solution helps to identify the maximum transfer capacity of tie-lines (Abbasi et al. 2021; Reddy 2019; El-Fergany and Hasanien 2020). The security appraisal helps to ascertain if the power system in a normal state can survive the contingencies (Zhang et al. 2019). The power system safety must be taken care of for smooth functioning of complex interconnected network (Naderi et al. 2019; Taher et al. 2019). The literature survey has revealed that not much attention has been put up on simultaneously solving the voltage-constrained OPF problem and optimal allocation of TCSC. This has motivated the authors to work in this direction.

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1.2 Recent research work

The FACTS devices are positioned to shrink the power flows in overloaded power lines, thereby providing a protective mechanism, bringing down power wastage and boosting the voltage consistency. Thyristor Controlled Series Compensator (TCSC) is a series-connected FACTS device that provides precise power control, damps inter-area oscillations, alleviate subsynchronous resonance and steps up transient stability.

Jordehi (2015) has proposed a brainstorm optimization algorithm to find the optimal position and control setting of static var compensators (SVC) and TCSC but the problem of OPF is not addressed. Subhojit Dawn et al. (Dawn and Tiwari 2016) presented a gradual change in TCSC regulating parameters and unified power flow controller (UPFC) for their allocation but the comparison of results with that obtained from hybrid approach is not carried out. Jordehi (2016) proposed an imperialistic competitive algorithm for the TCSC allocation problem that resulted in low voltage deviation and overloads during load growth and contingencies when compared to gravitational search algorithm (GSA) and bat swarm optimization but the generation cost optimization has not been carried out.

K. Pandiarajan and Babulal (2016) integrate fuzzy and harmony search algorithm to curtail the complete cost of generation fuel and obtained an OPF solution but both TCSC location and size not optimized. Naresh et al. (2016) employed an improved Harmony Search Algorithm for the coordinated TCSC design and power system stabilizers to effectively damp oscillations but TCSC allocation has not been proposed. The authors in Sekhar and Mohanty (2016) developed an upgraded cuckoo search algorithm for contingency constrained economic power dispatch aiming to relieve system overloading but FACT devices have not been incorporated. The authors in Abd-Elazim and Ali (2016) applied the cuckoo search (CS) algorithm for parameter tuning of STATCOM in a multi-machine system but optimization of generation cost has not been carried out. The authors in Nguyen (2016) proposed a CS algorithm for generating a solution to a combined heat and power load dispatch problem with a prime objective of fuel cost minimization but TCSC has not been incorporated. The cuckoo search algorithm (Nartu et al. 2019) is employed for reducing transmission losses by optimally placing FACT device but fuel generation cost not included. A novel hybrid algorithm (Dash et al. 2020a) exploited the properties of recent algorithms to minimize the losses and identify the location of TCSC and SVC in IEEE 14 bus and 30 bus systems but the OPF not considered. The installation cost was considered as the fitness function. Authors in Hocine and Djamel (2019) applied a genetic algorithm for sizing and placing UPFC. Several algorithms (Mahapatra

et al. 2019; Dash et al. 2020b; Abed-alguni and Alkhateeb 2020) emphasized cutting down of network power wastage and improving the node voltage profile simultaneously under normal and overloading instances. Salp swarm algorithm (SSA) is applied in Mahapatra et al. (2020) to determine the optimal reactive power planning in an interlinked network. The convergence is improved by coupling the Oppositional-based learning concept with the oppositional SSA and validated in a standard IEEE 57 benchmark system. Optimal power flow problems have been addressed in detail in Mahdad (2020), Nguyen 2019, Manusov et al. 2018) where single, as well as multiple objectives, are incorporated. The elimination mechanism is chosen over a selection mechanism to simplify selection efforts. The solutions with fast searchability were proposed and the objective functions were centred around the fuel cost, power loss, and voltage profile. In Raj et al. (2021), the authors applied salp swarm algorithm (SSA) and quasi-oppositional SSA for optimal TCSC placement in IEEE 14- and 57-bus system. In Failed (2021), the author utilized a modified branch-and-bound algorithm by application of depth-first search algorithm followed by breadth-first search algorithm in solution to the problem. In (Babu et al. 2020), the authors proposed a new voltage stability indicator in identifying the weakest bus. In Dey et al. (2020), author's hybrid combination of grey wolf optimizer, sine-cosine algorithm and crow search algorithm to minimize the generation cost. In Parihar and Malik (2017), the total system cost for generation is optimized by considering Economic load dispatch and Environmental Dispatch simultaneously.

1.3 Paper contribution

Currently, voltage constrained optimal power flow (VCOPF) problem is one of the most demanding problem and a crucial aspect to be considered for transmission expansion planning. It has been observed from the literature review that attempting simultaneous OPF solution with voltage consistency management and optimal FACTS controller allocation is an extremely challenging task. So, many of the existing algorithms individually may not produce best results when applied to complex power system optimization issues with multiple constraints. So, hybrid algorithms are most suited to tackle such complex optimization as they are designed and implemented for handling numerous large-scale, multimodal real-world problems. This research work utilizes the best attributes of CS-ALO hybrid algorithm in addressing the VCOPF problem. The summarized contribution of this paper is given as follows:

- Hybrid Cuckoo search (CS)-ant lion optimizer (ALO) algorithms are solicited to solve optimal locale and sizing of TCSC respectively.
- The independent approaches for the optimal position and TCSC size provide the versatility of the approach to the problem. The results can be independently utilized for further use.
- The results obtained are superior in terms of losses and TCSC cost when compared to the results in the published work. It also generates improved voltage profile.
- The outcomes signify that the recommended technique is feasible for implementation in large power networks with the incorporation of FACT devices.

1.4 Organization

Section 1 deals with the motivation to use a hybrid CS algorithm—Ant-Lion optimizer (ALO) to solve TCSC allocation in IEEE standard bus systems, literature review and the contribution of this paper. Section 2 covers the mathematical modelling of the OPF problem and TCSC allocation. Section 3 and 4 discuss the CS and ALO. Section 5 covers the solution methodology for TCSC allocation. Section 6 deals with results and discussion on two standard test systems. The conclusions are drawn finally.

2 Problem formulation and modelling of TCSC

OPF problem formulation aims at truncating the fuel cost which is the main component of generation cost. However, in the current work the authors also aim to enhance the system voltage profile of the complex interconnected network. So, the problem formulation in the work proposed is voltage constrained OPF solution with optimal TCSC placement.

2.1 Establishing the objectives of the current problem

The fuel cost component of the generation is illustrated in Eq. (1).

$$\text{Aggregated Fuel cost, } F_c = \sum (a_i + b_i P_{gi} + c_i P_{gi}^2) \$/hr \quad (1)$$

where a_i , b_i and c_i are the coefficients of fuel cost, P_{gi} represents real power produced by the i^{th} generator. The OPF is a constrained optimization problem. The active and reactive power balance would act as equality constraints

and the inequality constraints on the active power and voltage magnitude are given in Eqs. (4) and (5).

$$P_{inj,i} = P_{g,i} - P_{L,i} \quad (2)$$

$$Q_{inj,i} = Q_{g,i} - Q_{L,i} \quad (3)$$

$$P_{g,i}^{\min} \leq P_{g,i} \leq P_{g,i}^{\max} \quad (4)$$

$$V_i^{\min} \leq V_i \leq V_i^{\max} \quad (5)$$

In the paper, the optimal TCSC allocation is developed with the following objective:

$$OB_f = \text{Min}(\phi) \quad (6)$$

where, $\phi = \phi_1 + \phi_2 + \phi_3 + \phi_4$.

In the above equations, $\phi_1 = API$, $\phi_2 = V_{dev}$, $\phi_3 = P_L$ and $\phi_4 = C_{TCSC}$ are considered.

The apparent power flow index (API) is computed using Eq. (7) and has to be minimized.

$$API = \min \left(\frac{S_{i,Opf}}{S_{i,\max}} \left(\frac{S_{i,L} - S_{i,n}}{S_{i,L} - S_{i,opf}} \right) \right) \quad (7)$$

where, $S_{i,n}$ is the power flow on the i^{th} line, $S_{i,Opf}$ is the same power flow at OPF, $S_{i,\max}$ stands for the maximum power flow, and $S_{i,L}$ stands for the power flow at the time of the loading. The voltage deviation is calculated using Eq. (8).

$$V_D^i = \min \left(\sum_{i=1}^{NL} |V_i - V_i^j| \right) \quad (8)$$

where V_i represents the voltage magnitude at the i^{th} bus. The deviations in voltage are a result of system loading and revamping the voltage profile along with OPF solution is an uphill task. However, this is a requirement in an interconnected power network which is the objective of current work. The initial voltage value is set at 1 p.u. The power loss is computed using Eq. (9).

$$P_L = \sum_{j=1}^N |V_i| |V_j| |Y_{ij}| \cos(\theta_{ij} - \delta_i + \delta_j) \quad (9)$$

The TCSC cost (Habur 2002) is expressed in terms of TCSC operating range (S).

$$C_{TCSC} = 0.0015S^2 - 0.7130S + 153.75t) US\$/KVar \quad (10)$$

$$S = |Q_2 - Q_1| MVar \quad (11)$$

Where Q_2 and Q_1 are reactive power in the lines after and prior to TCSC installation.

2.2 TCSC modelling

A TCSC comprises of capacitor shunting anti-parallel SCRs and an inductor. The delay angle variation regulates the reactance of the thyristors comprising the TCSC. As the power delivery capability of a transmission line is inversely related to net reactance offered by the path inclusion of TCSC in series with the line diminishes the transfer reactance. This eventually leads to the enhancement of maximum power transfer of the line coupled with the reduction in reactive power loss. The modes of operation of TCSC may be inductive or capacitive and the level of compensation is guided by Eq. (12) to ignore over-compensation. A suggestive indication of TCSC inception in a power line is provided in Fig. 1.

The boundary limit of TCSC reactance is presented below:

$$-0.7X_{line} \leq X_{TCSC} \leq 0.2X_{line} \quad (12)$$

3 Cuckoo search algorithm

The bio-inspired Cuckoo Search algorithm (Yang and Deb 2009) is derived from the exceptional nature of the cuckoo bird which lay its eggs in another bird nests and may even damage the other eggs to make the survival of its eggs better. The superior nest is one which good eggs and will be moved to the next generation. The accessible nests are definitive and the probability of searching for an alien egg by the host bird is $p_a \in (0, 1)$. This ensures two eventualities to take place either the host bird identifying the alien egg such that it is destroyed, or a separate niche is created again as a nest (Joshi et al. 2017; Dubey et al. 2016). The available potential solution when it is presented as a mathematical approach are the eggs in the nest. The algorithm thus commences with an irregularly created population and solution set. To limit the random characteristics of the created population a D-dimension vector

that defines the parameter between extreme limits are as given below:

$$\delta_{\min} = \{\delta_1^{\min}, \delta_2^{\min}, \dots, \delta_D^{\min}\} \quad (13)$$

$$\delta_{\max} = \{\delta_1^{\max}, \delta_2^{\max}, \dots, \delta_D^{\max}\} \quad (14)$$

Thus, the j th component generated from the i^{th} vector is given as

$$\delta_{j,i}^t = \delta_j^{\min} + rand_{ij}[0, 1] \cdot (\delta_j^{\max} - \delta_j^{\min}) \quad (15)$$

where, $i = 1, \dots, NP$, $j = 1, \dots, D$ and $rand_{ij}[0, 1]$ function generates uniformly distributed random values in the interval $[0, 1]$.

A levy flight is performed during the generation of a new solution as in Eq. (16).

$$\delta_i^{t+1} = \gamma_i^t + \alpha \oplus Levy(\lambda) \quad (16)$$

The terms in δ_i^{t+1} depicts a novel solution, δ_i^t is the present location and $Levy(\lambda)$ is the probability of transition. The notation $\oplus$ represents product developed entry wise. A levy flight performance is depicted below

$$Levy \approx u = t^{-\lambda} (1 < \lambda < 3) \quad (17)$$

Based on the probability density function an arbitrary walk is engaged which is performed as an instantaneous hop (Yang and Deb 2009). This is done during every egg selection process and the objective function is shifted to the next position in case the next egg is a better solution in search space. This process of adopting levy flight method is to seal a fast convergence towards optimality. The preceding event in the CS is to build a new solution by selecting nests and is presented in Eq. (18).

$$\gamma = \begin{cases} \delta_i + rand \cdot (\delta_{r1} - \delta_{r2}) & rand < p_a \\ \delta_i & otherwise \end{cases} \quad (18)$$

The computed current solution γ_i is contrasted to δ_i . If the current solution is found to be better, then γ_i is accounted as a prime solution, else δ_i will be retained.

4 Ant lion optimizer

Ant Lion Optimizer (ALO) is a bio-inspired algorithm that mimics the predator and pasturing ability of antlions (Mirjalili 2015). The larva stage is dominated by foraging and is engaged in the reproduction cycle on maturity. The antlions are characterized by their strange circular movement to establish cone-shaped dugouts in the sand. These are halting station where they wait to capture their prey (ants). The pits size has a direct influence by moon size and the starvation level of the antlion. The cone-shaped structure of dugouts is staged in such a manner that it is

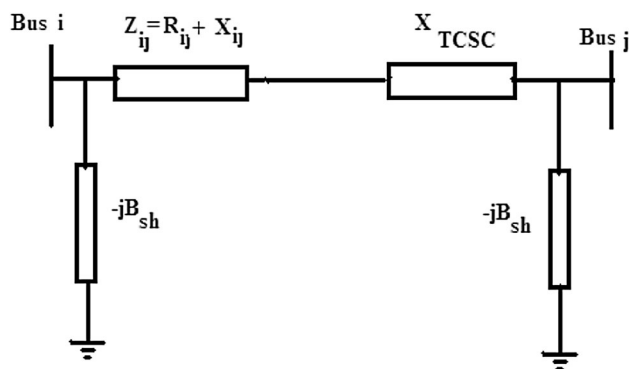


Fig. 1 Representation of TCSC inception in power line

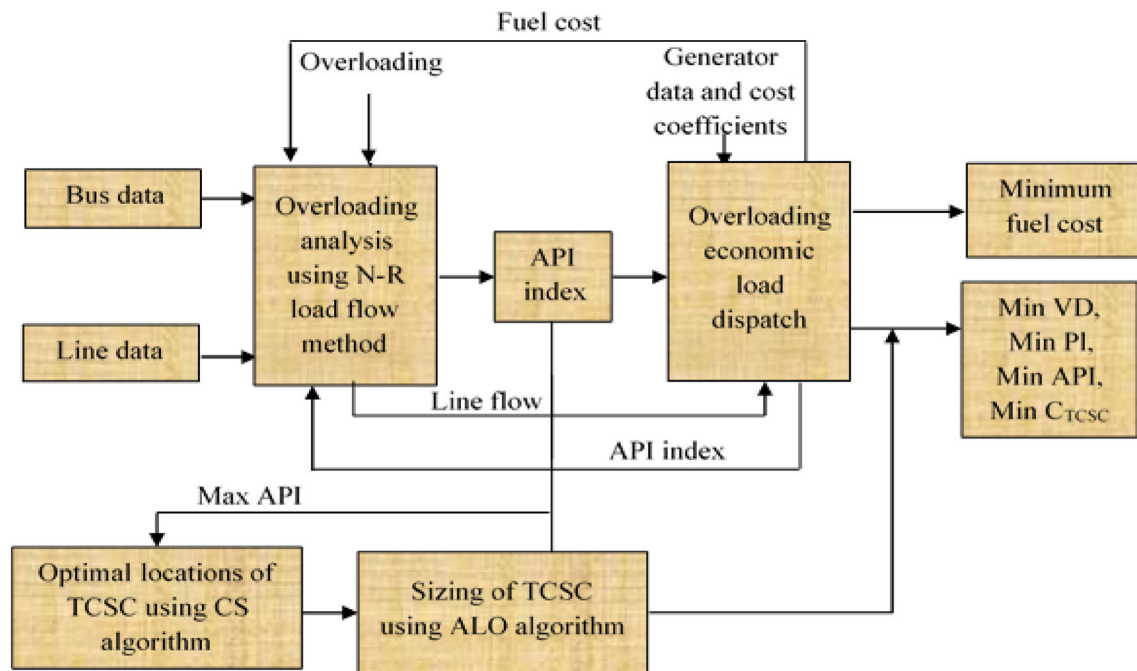


Fig. 2 Structure of the proposed method

slippery, and ants can easily be trapped into it. The ALO algorithm is characterized by five vital steps: arbitrary movement of ants, building trap pits, deceiving the ants and reconstruction of the dugouts for the next prey.

The ALO method is an adaptation of this interlinkage between antlion and the ants as predator and prey. Mathematically modelling this bio-inspired interdependency requires a stochastically driven approach. The ant matrix is randomly initialized and upgrade in position to the next level is decided by the roulette wheel operator and the elite. To avert excessive divergence in search space, the bounds are specified corresponding to the current iteration. The arbitrary movement of ants around selected elite and antlion is carried out until all the prey are evaluated and fitness is evaluated.

The advantages associated with the ALO technique are numerous such as the ability to reach global optima due to ample diversification in search space by arbitrary antlion selection, flexible contraction of antlions boundaries and strengthened probability to prevent local optima stagnation due to diversified population (Dubey et al. 2016).

5 Solution methodology for optimal TCSC allocation

In this work, the optimum TCSC setting, and sizing is combinedly established by using CS and ALO algorithm. The optimal allocation depends on the indices such as

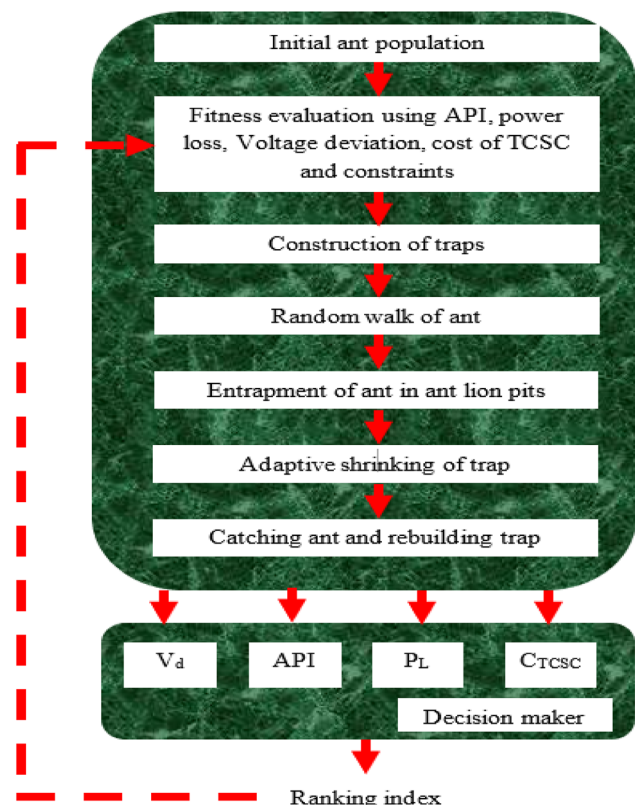


Fig. 3 Implementation of Antlion Optimization Algorithm in optimal sizing of TCSC

Apparent Power Flow (API), voltage variance (V_D), power dissipated (P_L) and cost of TCSC (C_{TCSC}). The best TCSC

location is governed by the CS algorithm. Subsequently, the injection capacity of the TCSC controller is determined by security-constrained ALO algorithm. The procedure adopted is candidly depicted in Fig. 2. The power flow is generated by analyzing the overloading conditions (Fig. 3). Thereafter, the considered power network is subjected to loaded fault condition to verify the robustness of the proposed methodology. Incorporation of TCSC restores system normalcy and reliable operation is ensured.

5.1 Optimal TCSC location using CS algorithm

The CS technique is solicited to generate TCSC optimal placement. The details of buses, power line and voltage are initialized for both the test systems. The Newton–Raphson algorithm is applied to compute power flows. As the system is overloaded, the bus voltages, losses and API values are computed. The optimum location is determined through the highest API. Thereafter, the fitness function is computed as in Eq. (19)

$$F_i = \max(API) \quad (19)$$

At the termination of the CS algorithm, precise solutions based on fuel cost reduction and foremost API value is obtained for TCSC optimal placement.

5.2 Optimal TCSC sizing using ALO algorithm

The ALO algorithm is executed for obtaining TCSC optimal cost solution as well as improving system overall voltage consistency. Here, the matrix M_{ant} saves the arbitrary ant's position and the fitness function merit of all ants are stored in the M_{OA} matrix. Similarly, the position and the fitness of antlion are represented in the $M_{Antlion}$, M_{OAL} matrices, respectively.

The application of the ALO algorithm is depicted in Eq. (20).

$$F_i = \min \begin{pmatrix} API \\ V_D \\ P_L \\ C_{TCSC} \end{pmatrix} \quad (20)$$

The antlions fitness is evaluated using the fitness function. Thereafter, the arbitrary ant's search for food is mathematically given as:

$$X(t) = [0, \text{cumsum}(2r(t_1) - 1), \text{cumsum}(2r(t_2) - 1), \dots, \text{cumsum}(2r(t_n) - 1)] \quad (21)$$

where cumsum is the cumulative sum, n is the maximum number of ants and t is the iteration. Also, $r(t)$ is a random number whose value is between $[0, 1]$. Min–max

normalization is followed to ensure faster convergence and prevent excessive randomness in the search space.

The trapping process of ants by antlion is mathematically modelled as follows:

$$c_m^t = Ant - lion_n^t + c^t \quad (22)$$

$$d_m^t = Ant - lion_n^t + d^t \quad (23)$$

In the process of building the trap, the roulette wheel method is applied for the selection of the fittest antlion. The sliding movement of ants into the hollow pits set by antlion is modelled as follows:

$$c^t = \frac{c^t}{I} \quad (24)$$

$$d^t = \frac{d^t}{I} \quad (25)$$

The parameters $I = 10^{w(\frac{t}{S})}$, and ' t ' is the present iteration while ' S ' signifies maximum iterations. ' w ' is a constant given below:

$$w = \begin{cases} 2 & \text{if } t > 0.1S \\ 3 & \text{if } t > 0.5S \\ 4 & \text{if } t > 0.75S \\ 5 & \text{if } t > 0.9S \\ 6 & \text{if } t > 0.95S \end{cases} \quad (26)$$

Antlion engulfs the ant as it falls into the dugout trap and then proceeds with a reconstruction process of the pit wherein the ant position update has to occur to engulf new prey.

$$Antlion_n^t = Ant_m^t \quad \text{if } f(Ant_m^t) > f(Antlion_j^t)$$

Elitism is applied to generate improved solution at every stage. The fittest antlion is given the status of elite (Parihar and Malik 2017) and it mobilizes the antlion arbitrary movement.

$$Ant^t = \frac{R_A^t + R_B^t}{2} \quad (27)$$

where, R_A and R_B represents the arbitrary walk throughout the designated antlion and elite at t th iteration, respectively. The culmination of iterative procedure occurs on optimization of the proposed objective function and the flow chart is presented below:

6 Results and discussion

In the current work, the CS and ALO algorithms are applied to evaluate the best siting and the capacity of the TCSC controller. The proposed methodology is also utilized to upgrade the voltage consistency at the buses with TCSC connected system. The work is implemented on

Table 1 Algorithm parameters

Algorithm	Description	values
CS	Maximum iterations	100
	Levy distribution factor (beta)	3/2
	Population size (N)	10
ALO	Maximum iterations	100
	Population size (N)	10
	$I = 10^{w(\frac{t}{T})}$ (From Eq. 26)	1

Table 2 Optimal Location and capacity of TCSC in IEEE 6-bus network

Best line	Bus start	Bus end	TCSC power (MVar)	Reactance X_{TCSC} p.u	Voltage variation (pu)	Power loss (MW)	TCSC Cost (\$/KVar)
10	4	5	0.079613	0.106	0.0013	3.0015	92.7839
11	5	6	0.090728	0.189	0.0311	3.0634	100.2259

Table 3 Real Power flows before and after TCSC placement in 6-bus network

Real power (MW) Basecase	Generator	Real power (MW)			
		F-GSA (Joshi et al. 2017)	RBFNN- GSA (Joshi et al. 2017)	IGSA-FA (Joshi et al. 2017)	Proposed CS-ALO
80.3723	G ₁	86.2560	88.0805	90.2153	90.31526
84.9135	G ₂	54.2368	58.1641	68.324	72.2654
63.4237	G ₆	59.3281	63.2622	64.0356	62.4154

Table 4 Comparative analysis for Voltage Constrained OPF solution for test system 1

Parameters	Uncompensated	F-GSA (Mahapatra et al. 2016)	RBFNN –GSA (Mahapatra et al. 2016)	IGSA-FA (Mahapatra et al. 2016)	Proposed CS-ALO
Real power (MW)	214.89	209.98	210	210	210
Power Loss (MW)	12.5393	5.6314	3.8315	3.4213	3.0634
Fuel cost (\$/hr)	3134.2432	3064.4285	3050.3739	3013.3921	2908.048
TCSC cost (\$/KVar)	–	153.8921	148.7653	119.0594	100.2259
Voltage deviation (pu)	–	0.1475	0.1248	0.0834	0.0311

IEEE 6- and 14-bus test system on the MATLAB platform and their performances are critically studied by comparing it with other techniques as presented in the literature. The details about buses, transmission lines and the control variables limits are considered from IEEE-14 bus transmission system data is refereed from xxxx; Duong et al. 2013). The Newton–Raphson method is applied to generate the load flow condition prior to TCSC inception in the power network. The implementation variables for the algorithms are given in Table 1.

6.1 Test system 1- IEEE 6 Bus power network

The maximum API identifies the best TCSC location in the given test system. Thereafter, the objective function is evaluated using the proposed approach. The TCSC is optimally placed in lines 4, 6, 12, 22 and 33 and its compensation level is set by the ALO algorithm. Table 2 provides the TCSC parameters used in the current work for voltage constrained OPF solution. The TCSC reactance is the most suited value to attain reactive compensation in the

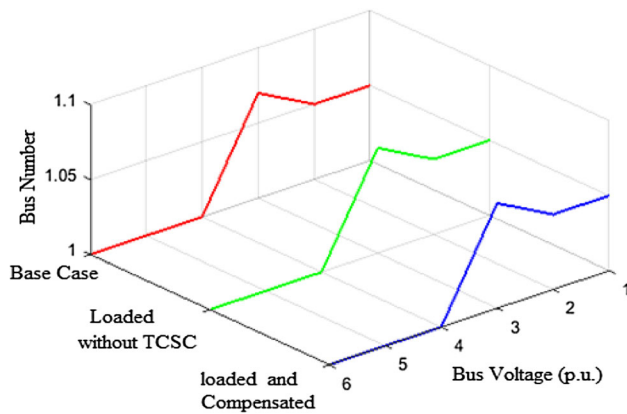


Fig. 4 Comparison Analysis of voltage profile under base case, uncompensated and TCSC compensated test system

test system. The corresponding value of TCSC power, voltage deviation and power loss are also provided in the table. Table 3 provides the active power generations in all generators of the test systems. The optimal TCSC placement relieves the system of overloading and enhances voltage profile improvement. The comparative analysis of the proposed CS-ALO method is provided with three other techniques for IEEE 6 bus test system. The proposed method is compared with the hybrid methods viz. Fuzzy-Gravitational Search Algorithm (F-GSA), Radial Basis Function Neural Network- GSA (RBFNN-GSA) and Improved Gravitational Search Algorithm- Firefly Algorithm (IGSA-FA), respectively.

The loading variation provides the required API value and the optimal placement of TCSC is in lines 10 and 11. It identifies and rectifies the voltage variation at the required bus and analyzes the system stability using the proposed method. The generated real power, real power losses,

voltage deviations, cost of TCSC and the generation cost of the system are evaluated and tabulated in Table 4. Figure 4 provides the voltage profile comparison under base case condition, uncompensated and TCSC compensated system. It can be observed from the graph that the voltage consistency is maintained during loading condition for a compensated system using the proposed CS-ALO technique. Figure 5 similarly provides the power loss variation under the base case and for the uncompensated and compensated system, respectively. It can be observed that there is a surge in power wastage when the network is loaded, and it is minimized by optimal TCSC placement using the proposed technique. This candidly depicts the efficacy of the proposed hybrid algorithms.

6.2 Test System 2- IEEE 14 bus

The efficacy of the proposed algorithm is also evaluated on IEEE 14 power network. The power flow computed using N-R method with the base case loading condition and to ascertain the efficacy of the CS-ALO algorithm load of the selected line is increased from the base case loading condition when power flow limits are considered. The secured OPF with a consistent voltage profile is established by optimal TCSC placement which alleviates the overload in transmission lines. Table 5 provides the TCSC parameters when placed at the optimal location as tabulated below.

Table 6 provides the real power generation schedule and Table 7 validates the performance of the recommended methodology with three methods published in the literature. The best generation scheduling is aptly maintained even under loaded condition by using the proposed method. The total real power losses and the generation cost of the systems are evaluated with and without TCSC in Table 7.

Fig. 5 Comparison analysis of power loss under base condition, uncompensated and TCSC compensated test system

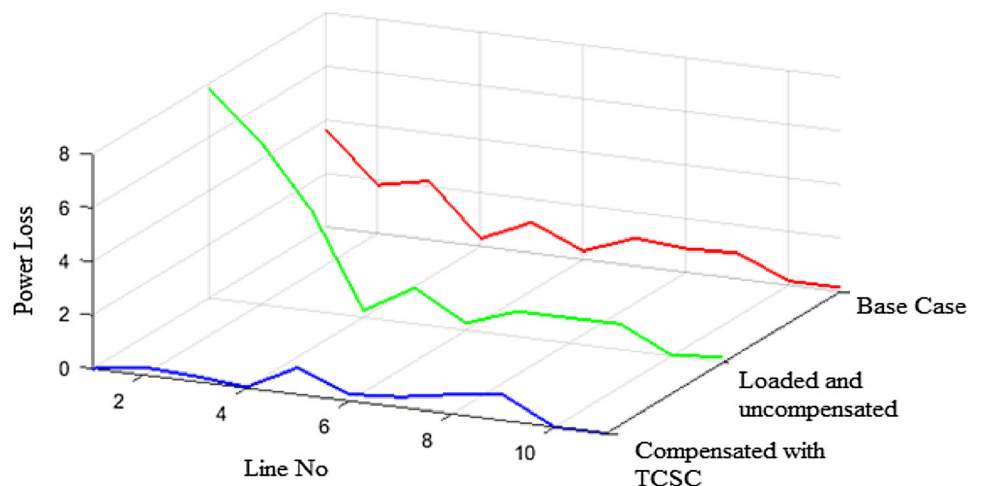


Table 5 TCSC best placement and capacity using proposed method

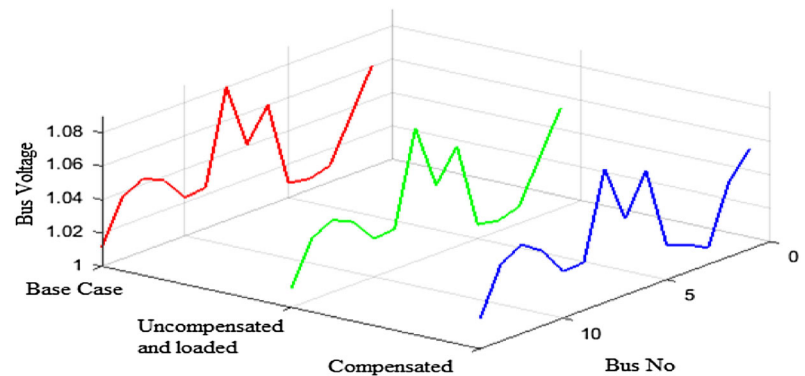
Best line	Start bus	End bus	TCSC capacity MVar	Reactance p.u	Voltage variation p.u	Power loss in MW	Cost of TCSC \$/KVar
8	4	7	0.191891	-0.2091	0.2125	9.2976	121.54646
15	7	9	0.19192	0.11	0.2158	10.1753	125.7454
6	3	4	0.1855	0.171	0.1382	9.6714	120.8972

Table 6 Real Power Generation profile for IEEE-14 power network

Base load (MW)	Generator	Real power (MW)			
		F-GSA (Mahapatra et al. 2016)	RBFNN- GSA (Mahapatra et al. 2016)	IGSA-FA (Mahapatra et al. 2016)	Proposed CS-ALO
141.7	G ₁	141.8538	150.54	139.6159	140.6541
79.93441	G ₂	71.3896	53.38314	71.60923	72.3254
53.58641	G ₆	46.7304	54.88227	59.74847	55.2354

Table 7 Voltage Constrained OPF profile and performance analysis of CS-ALO

Parameters	Without TCSC	F-GSA (Joshi et al. 2017)	RBFNN- GSA (Joshi et al. 2017)	IGSA-FA (Joshi et al. 2017)	Proposed CS-ALO
Real power (MW)	259	259.9539	259	258.4563	258
Power Loss (MW)	13.5929	7.6000	6.8227	4.6322	4.2976
Fuel cost (\$/hr)	3460.2349	3125.9924	3050.1011	1297.4326	1156.8628
TCSC cost (\$/KVar)	145.3623	132.6382	117.5868	100.2259	
Voltage deviation (pu)	0.5147	0.4834	0.0728	0.0311	

Fig. 6 Voltage profile under basecase, uncompensated and TCSC compensated IEEE 14 network

The proposed method has generated improved results as compared to the other techniques for diminishing the fuel cost, real power loss, TCSC cost and voltage deviation. The reduction in fuel cost with the projected method is 10.834% and power loss by 7.2235% when compared to the IGSA-FA method.

Figure 6 provides the voltage profile improvement with optimal TCSC location in the test system. It showed that the recommended method constructively maintains the voltage consistency within the prescribed limit (1.02–1.0 p.u.) when the test system is loaded and compensated with a TCSC controller. Figure 7 depicts the power loss for the three cases of base loading, loaded and uncompensated

Fig. 7 Real power loss under base condition, uncompensated and TCSC compensated IEEE 14 network

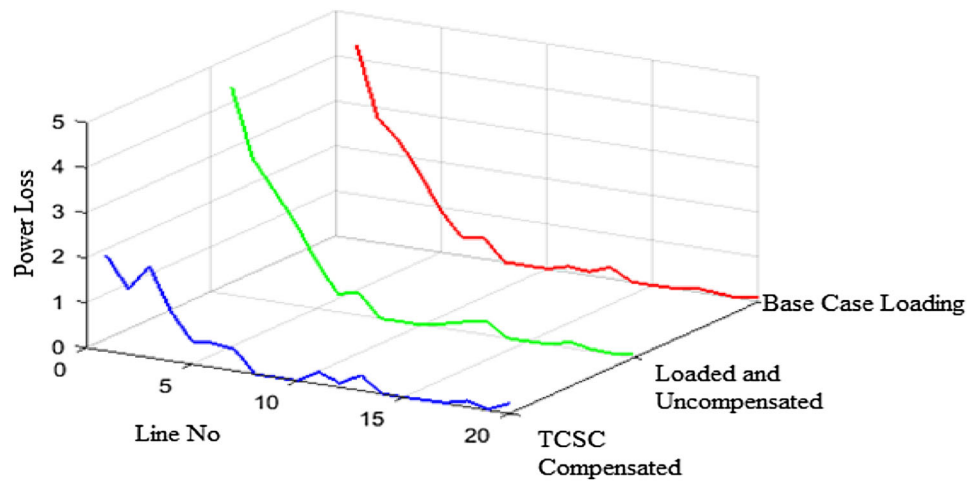
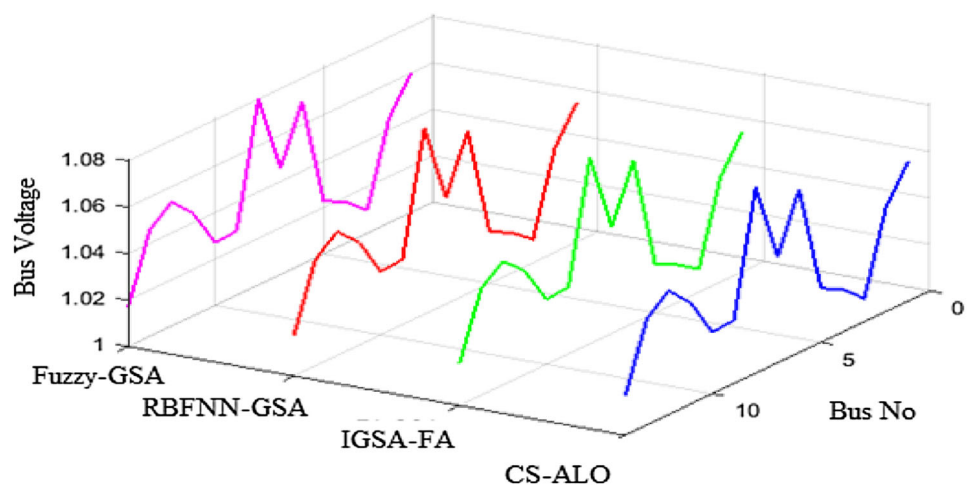


Fig. 8 Comparison of voltage profile under basecase, uncompensated and TCSC compensated IEEE 14 network



system and TCSC compensated system. The diminished power loss with the implementation of the proposed technique with optimal TCSC placement is candidly observed in the graph. Similarly, Fig. 8 provides the voltage profile variation of the recommended algorithm contrasted with the other techniques. Figure 9 provides a comparison between all the algorithms concerning power loss and it is observed that the active power loss has been significantly reduced using the proposed hybrid algorithms. So, it is evident that the test system is secured with optimal TCSC placement with no overloading and improved power transfer capability and voltage profile of the system.

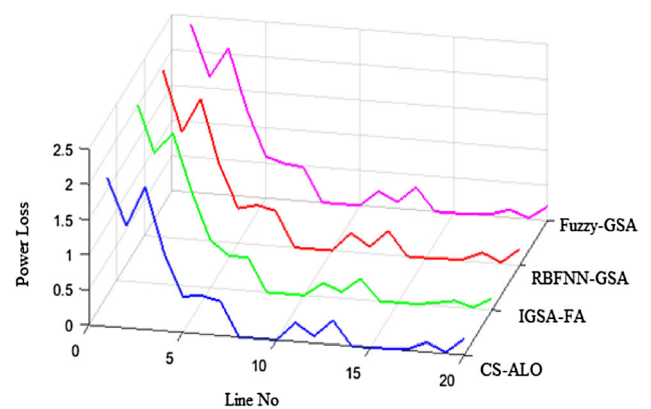


Fig. 9 Voltage profile for various algorithms in a TCSC compensated IEEE 14 network

Table 8 Statistical analysis of Proposed CS-ALO Algorithm

Best TCSC placement at Lines	Statistical values	Voltage variation (p.u.)	Power loss (MW)	Cost of TCSC (\$/KVar)
10	Mean	0.5131	4.1078	105.0854
	SD	0.7106	0.6052	1.4293
	Min	0.0311	3.0634	100.2259
	Max	1.915	5.1989	105.0854
11	Mean	0.7767	4.393	102.7696
	SD	0.4425	0.8561	1.3887
	Min	0.0311	3.0015	92.7839
	Max	1.5112	6.0069	105.0881

Table 9 Statistical analysis of proposed method

Best TCSC placement at lines	Parameters	Voltage variation (pu)	Power loss (MW)	Cost of TCSC (\$/KVar)
8	Mean	0.0987	10.7302	123.9257
	SD	0.0608	0.5686	2.2834
	Min	0.0708	9.2976	121.5465
	Max	0.1952	11.254	127.882
15	Mean	0.1472	10.3589	123.1618
	SD	0.0849	0.532	1.9925
	Min	0.2158	10.1753	125.74
	Max	0.29	11.2392	130.9988
6	Mean	0.1241	10.6709	124.2435
	SD	0.0689	0.9535	2.2769
	Min	0.1382	9.6714	120.8972
	Max	0.2449	12.24	127.9804

6.3 Statistical Analysis

The statistical analysis is carried out in terms of average value, standard deviation (SD), maximum and minimum values of voltage magnitude, power losses and TCSC cost. This further validates the efficacy of the proposed method in achieving superior performance in the current power system problem. Table 8 and Table 9 provides the statistical analysis for the proposed CS-ALO in the current problem for IEEE 6 and IEEE 14 network.

The execution of the considered method is examined in terms of two statistical measurements viz. individual relative error (IRE) and individual absolute error (IAE). They are depicted as follows:

$$IAE = \frac{TCSC \text{ with other method} - TCSC \text{ with Proposed Method}}{TCSC \text{ with Proposed Method}} \quad (29)$$

Figures 10 and 11 presents the contrast among various algorithms graphically through IRE calculation with respect to various objectives considered in the present work. It shows that the recommended CS-ALO method has outperformed the other techniques in all criteria that have been accounted for.

Similarly, Figs. 12, 13 and 14 have shown the relative performance of the CS-ALO algorithm with the other methods. The proposed technique has substantially achieved enhanced voltage constrained OPF solution by optimal TCSC placement.

$$IRE = \frac{(TCSC \text{ with other method} - TCSC \text{ with Proposed Method})}{TCSC \text{ with Proposed Method}} \% \quad (28)$$

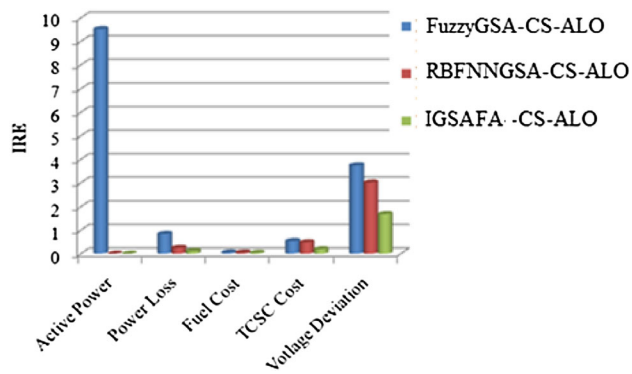


Fig. 10 Comparison analysis by IRE for test system 1

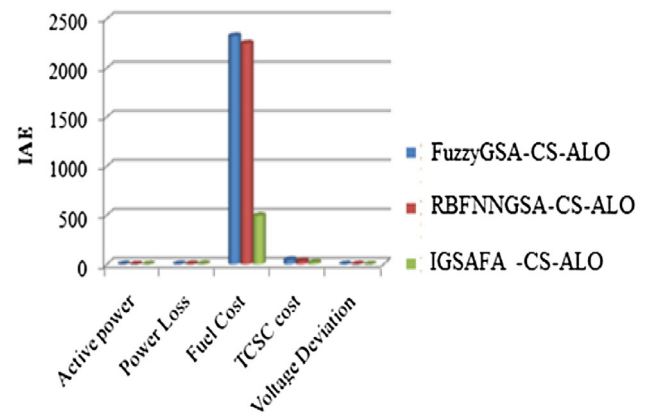


Fig. 13 Comparison analysis by IAE for test system 2

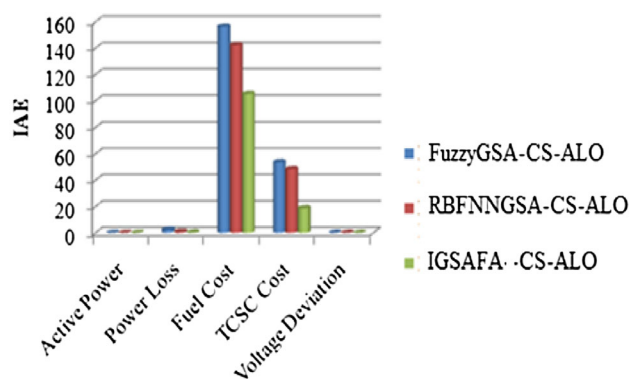


Fig. 11 Comparison analysis by IRE for test system 2

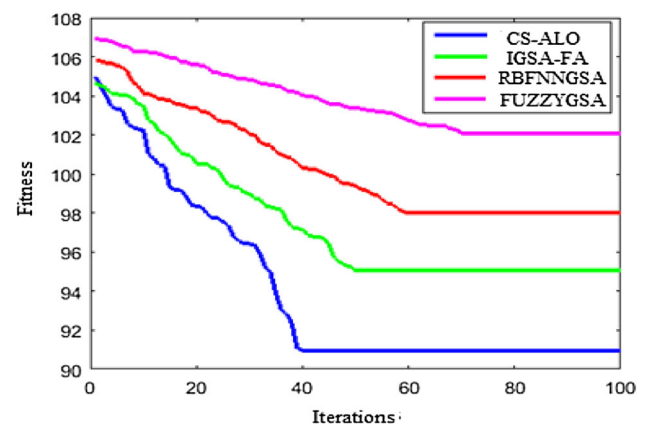


Fig. 14 Convergence curve for test system 1

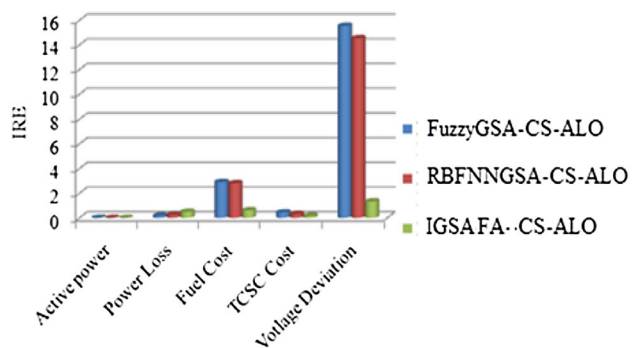


Fig. 12 Comparison analysis by IAE for test system 1

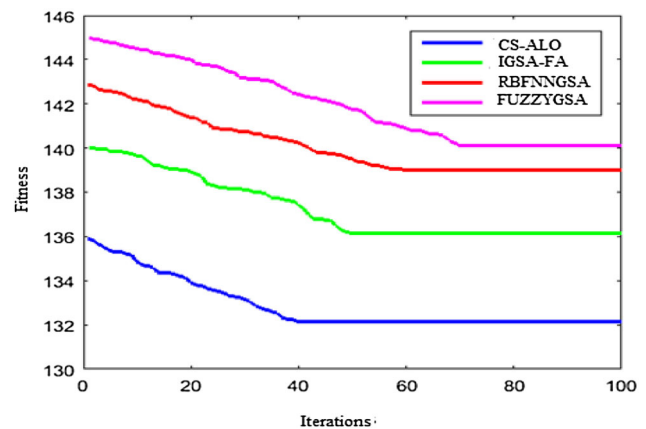


Fig. 15 Convergence curve for test system 2

Figures 14 and 15 illustrates the convergence curves for all the algorithms considering IEEE 6 bus and 14 bus, respectively. The convergence criteria have been established over 100 iterations and from both the graph the proposed CS-ALO technique has been found to depict

superlative performance. The quick convergence signifies its ability to have more computational efficiency when utilized in a complex optimization issue.

7 Conclusion and future scope

The optimal TCSC location and size is identified by the Cuckoo search algorithm and ant-lion optimizer, respectively. They helped in resolving voltage-constrained optimal power flow issue in IEEE 6 and 14 bus power network. The quadratic fuel cost was chosen as the objective. The outcomes signified a substantial reduction in reactive and real loss of power and voltage deviations in normal and overloading conditions. The power flow security is restored eventually after the overloading conditions through the use of TCSC. The results achieved with the proposed algorithms are compared to three other hybrid techniques and found to be superior in all respect. The voltage profile has improved in all the test cases. The statistical analysis and convergence characteristics substantiate the robustness of the proposed method. This work can be extended to larger bus system. It can be concluded that the proposed hybrid CS-ALO algorithm provides more accuracy and reliability to obtain global or near global optimal settings of control variables. The effective implementation of the proposed algorithm in the given engineering application ensures that in future it can also be extended to large size interconnected power networks and other complex power system optimization issues like Economic load dispatch, power system stability issues etc.

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Declarations

Conflict of interest All Authors declare that they have no conflict of interest.

Informed consent: Informed consent was obtained from all individual participants included in the study.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent: Informed consent was obtained from all individual participants included in the study.

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